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Math 402/436

Introduction: The Prediction Problem

Setup. A **return** is the daily percent change in a stock's price: $r_{i,t} = (P_{i,t} - P_{i,t-1})/P_{i,t-1}$. A **signal** $z_{i,t}$ is a quantitative predictor (cross-sectionally z-scored). We want to estimate $\mathbb{E}[r_{i,t+1} \mid z_{i,t}]$ to build a long-short portfolio.

Classical answer (Grinold & Kahn, 1999). If $(z, r/\sigma)$ is bivariate normal with correlation IC, conditioning gives:

$$\underbrace{\mathbb{E}[r_{i,t+1} \mid z_{i,t}]}_{\text{alpha}} = \mathrm{IC} \cdot \sigma_{i,t} \cdot z_{i,t},$$

where $\sigma_{i,t}$ is predicted idiosyncratic volatility and $\mathrm{IC} \approx 0.05$ is treated as a constant.

Our question. Does IC vary with signal strength z and over time t? Can we estimate $\mathrm{IC}_t(z)$ to improve on the static baseline?

Two structural reasons to try: (1) Some signals have persistently negative IC—the static $+0.05$ is structurally wrong. (2) The IC function is non-linear in z (the data will show this).

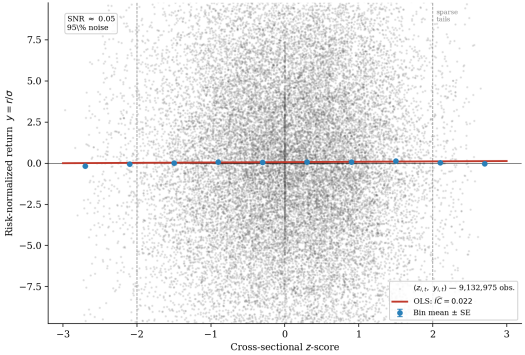
Honest caveat. $\mathrm{SNR} \approx 0.05$ means 95% of return variance is noise. We track slow structural drift over months—not rapid day-to-day regime detection.

Data and Feature Engineering

Dataset. Barra U.S. equity factor model, 2010–2024. $\approx 3{,}000$ stocks/day, 8 daily signals: style/industry/idiosyncratic momentum & reversal, idiosyncratic volatility, BAB. Reversal signals are negated at construction (high recent loss $\Rightarrow$ high z).

Risk normalization. Regress on $y_{i,t} = r_{i,t+1}/\sigma_{i,t}$ instead of raw returns. Under the Grinold model $\mathbb{E}[y_{i,t} \mid z_{i,t}] = \mathrm{IC}_t(z_{i,t}) \cdot z_{i,t}$, so the OLS slope is directly IC, free of cross-sectional scale differences.

Other preprocessing. Winsorize signals to 1st–99th percentile; clip z-scores to $[-3, 3]$; price filter $> \$5$ to remove illiquid micro-caps.



Each dot: one $(z_{i,t}, y_{i,t})$ pair over a 120-day window ($\approx$360k obs). The OLS slope (IC ≈ 0.05) is nearly horizontal against the noise cloud. **Blue circles** (bin means) show IC is non-constant in z—motivating a nonlinear estimator. Observations thin past $|z| = 2$: tail IC estimates are least reliable.

Methods: Kernel Ridge Regression

Goal. Learn $f : \mathbb{R} \to \mathbb{R}$ (z-score $\mapsto$ IC estimate) from the rolling 120-day buffer of (z_i, y_i) pairs.

Regularized least squares in the RKHS (RBF kernel $k(z, z') = C \exp(-(z - z')^2/2\ell^2)$):

$$\min_{f \in \mathcal{H}} \sum_{i=1}^{n} (f(z_i) - y_i)^2 + \alpha \|f\|_{\mathcal{H}}^2.$$

The **Representer Theorem** says any minimizer has the form $f(z) = \sum_i c_i k(z, z_i)$. Substituting into the objective and letting $\mathbf{K}_{ij} = k(z_i, z_j)$:

$$c = (\mathbf{K} + \alpha \mathbf{I})^{-1} y, \qquad \hat{f}(z^*) = k_*^\top (\mathbf{K} + \alpha \mathbf{I})^{-1} y.$$

SNR calibration. With $\mathrm{SNR} = 0.05$, the noise-to-signal ratio is $1/\mathrm{SNR} = 19$. We fix $\alpha = 19$: unless the data overwhelmingly support a non-zero IC, the prediction is shrunk toward zero.

Key property. $\hat{f}(z^*)$ is unconstrained in sign $\Rightarrow$ model can output $\mathrm{IC} < 0$ when the data support it.

Methods: Gaussian Process Regression

A GP places a prior directly over functions. $f \sim \mathcal{GP}(0, k)$ means every finite collection $(f(z_1), \ldots, f(z_n))$ is jointly Gaussian with covariance $\mathbf{K}_{ij} = k(z_i, z_j)$. We use the zero-mean prior: without data, the IC is expected to be near zero.

Observation model. $y_i = f(z_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$. The joint distribution of y and a test point $f_* = f(z^*)$ is Gaussian; conditioning on y gives the exact posterior:

$$f_* \mid y \;\sim\; \mathcal{N}\left(\underbrace{k_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} y}_{\mu_*(z^*)},\; \underbrace{k(z^*, z^*) - k_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} k_*}_{\sigma_*^2(z^*)} \right).$$

Two design choices. (1) Fix $\sigma_n^2 = 0.952$ (noise proportion at $\mathrm{SNR} = 0.05$)—prevents the optimizer from collapsing $\sigma_n^2 \to 0$ and interpolating through noise. (2) The per-stock alpha variance $\mathrm{Var}[\hat{\alpha}_{i,t}] = z_{i,t}^2 \cdot \sigma_*^2(z_{i,t})$ is added to the portfolio covariance matrix, down-weighting stocks where the IC estimate is unreliable.

Note. $\mu_*(z^*)$ is identical to the KRR prediction with $\alpha = \sigma_n^2$. GPR's advantage is the exact posterior variance.

Methods: Portfolio Construction

Alpha forecast. Both models output $\widehat{\mathrm{IC}}_t(z)$; the alpha for stock i is:

$$\hat{\alpha}_{i,t} = \widehat{\mathrm{IC}}_t(z_{i,t}) \cdot \sigma_{i,t} \cdot z_{i,t}.$$

Mean-variance optimization (MVO). Given alphas μ_t and Barra factor covariance Σ_t, solve:

$$\max_{w \in \mathbb{R}^N} \; w^\top \mu_t - \tfrac{\gamma}{2} w^\top \Sigma_t w \quad \text{s.t.} \quad w^\top \beta_{\mathrm{mkt}} = 0, \quad \mathbf{1}^\top w = 0,$$

with risk aversion $\gamma = 200$. Constraints enforce **market-neutrality** (zero market beta) and **dollar-neutrality** (equal long/short), so returns reflect pure signal skill. The unconstrained solution $w^* = \frac{1}{\gamma}\Sigma_t^{-1}\mu_t$ is projected onto the feasible set via Lagrange multipliers (implemented in CVXPY).

Walk-forward discipline. On each day t: update the 120-day buffer with yesterday's (z, y) pairs $\rightarrow$ refit $\rightarrow$ predict $\rightarrow$ optimize $\rightarrow$ trade. No look-ahead bias: the model at time t never sees data from $t+1$ onward. First 120 days are burn-in.

Results: Equal-Weighted Portfolio (All 8 Signals)

Model	Ann. Return %	Vol %	Sharpe	Max Drawdown %
Kalman Poly	0.94	0.79	**1.189**	−0.73
Static	1.01	1.00	1.006	−2.69
GPR	0.90	0.92	0.979	−0.87
KRR	0.54	0.88	0.613	−1.24

All four models produce positive Sharpe ratios on a fully market- and dollar-neutral portfolio. The static baseline is competitive—dynamic IC estimation does not dominate at the portfolio level.

Why? Diversifying across 8 signals averages out the cases where dynamic IC helps (sign-uncertain reversal signals) with the cases where it hurts (momentum signals whose IC is persistently positive in this sample).

Results: Where Dynamic IC Helps and Hurts

Signal / Model	Ret%	Vol%	Sharpe	MaxDD%
Idiosyncratic Reversal — dynamic models reduce drawdown dramatically				
Static	6.82	6.80	1.003	−17.57
GPR	5.05	3.99	**1.266**	**−1.62**
KRR	3.20	2.66	1.202	−1.32
Style Reversal — dynamic models rescue a negative-Sharpe static				
Static	−0.46	0.84	−0.550	−3.07
GPR	0.35	1.40	0.248	−2.29
Kalman Poly	0.69	1.15	**0.597**	−1.34
Idiosyncratic Momentum — static dominates; dynamic models add noise				
Static	1.55	1.30	**1.192**	−1.61
GPR	−0.08	1.19	−0.066	−2.79
KRR	0.03	1.06	0.028	−2.45

Dynamic models are most useful when IC sign is *structurally uncertain*. When IC is persistently positive, the static model's $+0.05$ is right and dynamic estimation adds noise at SNR $= 0.05$.

Ethical Implications

1. Crowding and self-defeating feedback.
If many institutions simultaneously estimate $\mathrm{IC}_t(z)$ and tilt toward high-$|z|$ stocks, correlated demand compresses the very premium being exploited. The sign-inversion feature amplifies this: simultaneous position reversals by many traders could trigger cascades. Position limits and turnover controls are necessary complements.

2. Access inequality.
These models require licensed Barra factor data, daily-frequency returns, and significant compute—resources unavailable to retail investors. Widespread institutional adoption could widen the information gap between professional and retail participants, effectively transferring wealth at the times when IC is most predictable.

3. Model fragility.
Both models assume stationarity within the 120-day buffer. A model that has learned $\mathrm{IC} < 0$ for a reversal signal may keep trading the inverted position after the regime has reverted, generating large drawdowns precisely when volatility is highest. Ongoing IC monitoring and automatic de-risking rules are essential.

Conclusions

Central finding

$\mathrm{IC}_t(z)$ is non-constant in both z and t. Kernel-based estimators can represent this, but the benefit is signal-specific, not universal.

Risk normalization	removes heteroskedasticity; slope = IC directly
SNR-calibrated regularization	$\alpha = 19$ (KRR), $\sigma_n^2 = 0.952$ (GPR)
Sign-flexible IC	both kernel models can output $\mathrm{IC} < 0$
Uncertainty propagation	GPR posterior variance enters MVO covariance
Rolling window	slow structural adaptation; not rapid regime detection

Bottom line. Dynamic IC estimation is most valuable for signals whose sign is structurally uncertain (e.g., reversal signals). When a signal's IC is persistently positive, the static $\mathrm{IC} = 0.05$ is hard to beat in a $\mathrm{SNR} = 0.05$ environment—the noise floor limits what any adaptive estimator can learn.